MIL-M-45131A(MU)
29 OCTOBER 1963

SUPERSEDING
MIL-M-45l31(ORD)
26 JUNE 1958.

MILITARY SPECIFICATION

MOUNT, PERISCOPE: M105

1. SCOPE

1.1 This specification covers one type of periscope mount known as Mount, Periscope: M105.

2. APPLlCABLE DOCUMENTS

2.1 The following documents, of this issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-F-13926 —Fire Control Materiel: General Specification Governing the Manufacture and Inspection of.

MIL-P-14232 —Parts, Equipment and Tools for Ordnance Materiel; Packaging of.

MIL-1-46208 —Inspection Requirements, General Specification for.

STANDARDS

MILITARY

MIL-STD-105 —Sampling Procedures and Tables for inspection by Attributes.

MIL-STD-109 — Quality Assurance Terms and Definitions.

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

F8237726 — Mount, Periscope: M105

LSQAP-8237726 — Supplementary Quality Assurance Provisions for Mount, Periscope: MI05

IEL-8237726 — List of Inspection Equipment Numbers for Mount, Periscope: M105

PACKAGING DATA SHEET

MIL-P-14232/P8237726 - Packaging of Mount, Periscope: M105

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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30 REQUIREMENTS

3.1 Materials. Materials shall be in accordance with drawings and specifications forming a part of this specification.

3.2 Fabrication. The mount shall be manufactured in accordance with Drawing F8237726 and drawings pertaining thereto.

3.3 General specification. The mount shall meet the following requirements of Specification MIL-F-13926:

(a) Order of precedence .
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.4 Vibration. The mount shall be vibrated at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch maximum excursion) for a period of 5 + 1/4 minutes. Subsequent to vibration the mount shall show no evidence of loose or damaged parts and shall meet the requirements of 3.6.

3.5 Orientation. The mount shall be positioned as shown in Figure 1 with the horizontal mounting surface in a horizontal plane within ± 10 seconds of arc.

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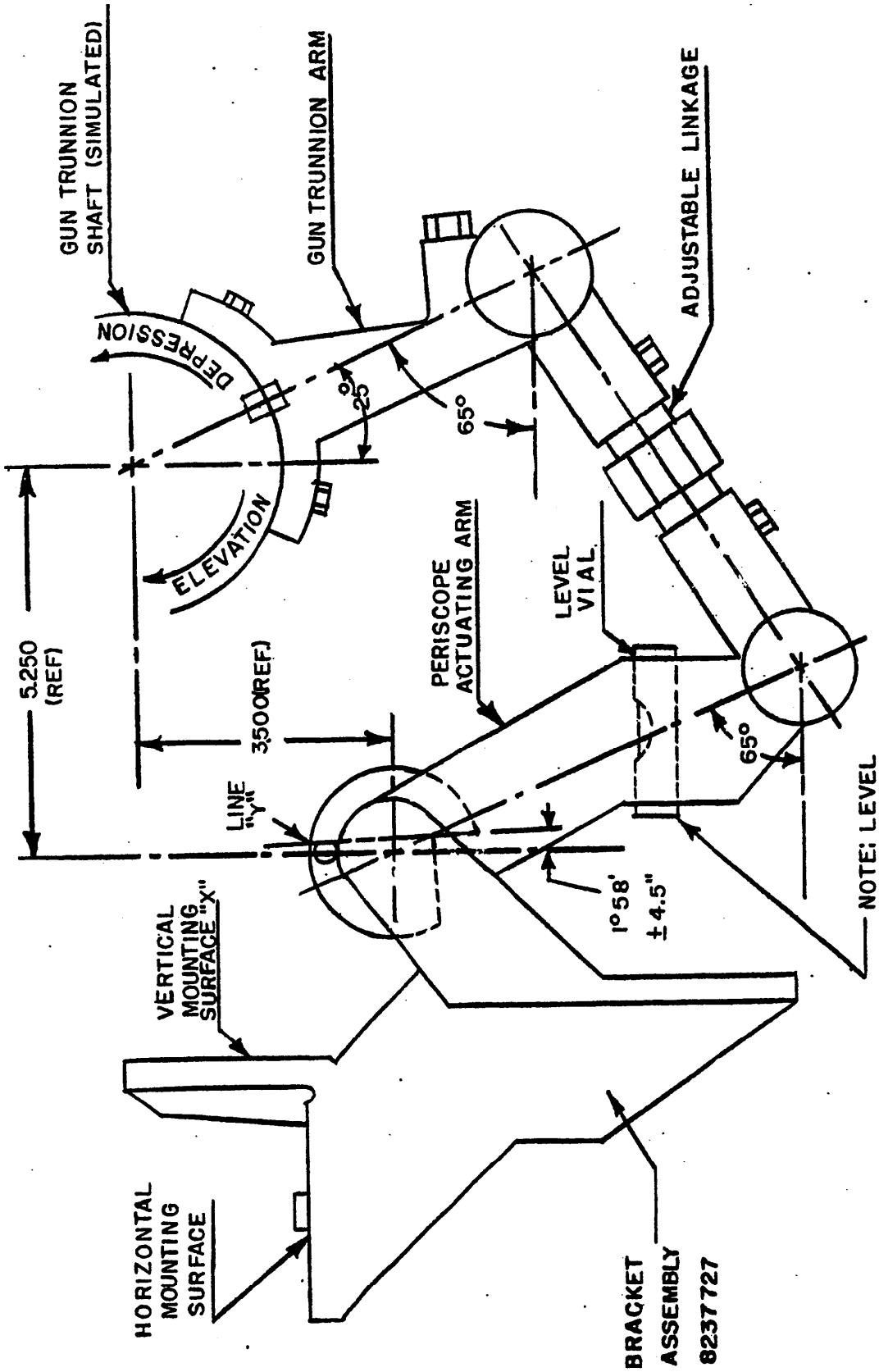


FIGURE 1. *Mount, periscope: M105.*

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3.6 Accuracy.

3.6.1 Output coupling. The simulated gun trunnion shaft shall be positioned in accordance with figure 1 and the linkage adjusted to center the level bubble. The output coupling shall be positioned as shown in figure 1 within the tolerance specified.

3.6.2 Elevation. With the mount positioned in accordance with 3.6, rotation of the simulated gun trunnion to a 120 depression and 22° elevation shall produce the same angles of rotation respectively at the output coupling within +0.2 mil, exclusive of backlash.

3.6.3 Backlash. Backlash between the output coupling and the simulated gun trunnion shaft shall not exceed 0.1 mil.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection shall be in accordance with the requirements of Specification MIL-1-45208 and the component inspection requirements of Specification MIL-F-13926. Definitions of inspection terms not otherwise defined herein shall be as listed in Standard MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Inspection lot. Unless otherwise specified by the contracting officer, inspection lot size formation, and presentation of lots shall be in accordance with "Submission of Product" specified in Standard MIL STD-105.

4.2.2 Examination and tests. Components and subassemblies shall be inspected in accordance with the inspection provisions contained in Specification MIL=F-13926 and supplementary Quality Assurance Provisions LSQAP-8237726. Examinations and tests related to section 8 herein shall be performed on a defect (individual characteristic) basis in accordance with Standard MIL-STD-105, and the inspection level and sampling plans specified in table I. Examinations and tests for Packaging, Packing and Marking shall be in accordance with Specification MIL-P-14232 and section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual nonconforming items.

TABLE I. Classification of defects

Use Level II of Table I with Sampling Plan Table II-A of Standard MIL-STD-105

Requiewmwnr test procesure		
CRITICAL: None.		
MAJOR: AQL 1.0%		
101. Physical failure, post vibration.	3.4	4.5.1.1
102. Output coupling	3.6.1	4.5,3
108. Elevation	3.6.2	4.5.4
104. Backlash	3.6.3	4.6.5
MINOR: None:		

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"*-4.2.3 Disposition of" nonconforming product. " Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in Standard MIL-STD-105.

4.3 Control tests. One mount shall be selected at random from each 100 produced, or from each months production, whichever occurs first. Control samples shall have successfully met all acceptance tests specified herein prior to conducting the following examinations and tests:

Control test	Requirement	Test procedure
Materials	3.1	Visual—Per Drawings and Specifications
Fabrication	3.2	Visual—Drawings
General Specifications	3.3	Visual—MIL-F-13926

4.3.1 Control test failure. Should the control test sample fail to meet the specified test requirements acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved.

4.4 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall furnish and maintain all required measuring and testing equipment in accordance with the "Inspection Equipment" requirement of Specification MIL I-45208. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Government furnished inspection equipment. Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with "Inspection Equipment," and *'Government Furnished Material" requirements of Specification MIL-I-45208.

4.4.2 Contractor furnished inspection equipment.

4.4.2.1 Government design. All inspection equipment required by drawings forming a part of the contract and not provided by the Government shall be supplied by the contractor in accordance with the List of Inspection E q u i p m e n t Numbers, IEL 8237726. Alternate designs may be substituted only as provided by "Inspection Provi sions" of Specification MIL-I-45208.

4.4.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5, and with the "Component Inspection" requirements of Specification MIL-F-13926. Concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment,

5 est methods nd procedures.

4.5.1 Vibration. Standard vibrating equipment capable of providing the amplitude and frequency specified in 3.4 shall be utilized for this test. The mount shall be vibrated in a vertical plane. Unless otherwise specified, the test equipment shall be in conformance with the test facilities requirement of Specification MIL-F-13926.

4.5.1.1 Post vibrations. Subsequent to vibration the mount shall be thoroughly examined for physical defects resulting from workmanship and materials to determine compliance with 3.4 prior to being subjected to the remaining tests herein. A visual examination shall be made for missing or

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damaged parts prior to being subjected to the remaining performance tests.- Minimum visual. examination shall include the level vial (also use smell test to detect leaking vial resulting from broken seal or glass), headrest, bracket and guard etc. A tactile examination should also be conducted to determine looseness of the above items and also the bracket assembly, linkage, etc. Any of the possible defects above shall be cause to consider a mount as "defective unless separate defects-per-hundred type sampling plans are established and approved by the Government as part of the suppliers inspection system. Mounts which have passed the above examinations shall be sub. jetted to the tests of 4.5.3 to 4.5.5 inclusive to determine overall compliance with 3.4

4.5.2 Orientation. When the mount is positioned in special test fixtures F-8299870 and F'-8299880 in accordance with the setup and operating instructions, the mount shall comply with figure 1 and 3.6

4.5.3 Output coupling. The output coupling will be positioned as specified in 3.6.1 and figure 1 when the mount is oriented in accordance with 4.6.2.

4.5.4 Elevation. With the mount positioned as specified in 4.5.2, revolve the simulated gun trunnion shaft to 12° depression from the zero position and 22° elevation from the zero position while viewing through the collimator. The error if any, when read from the collimator reticle shall be within the tolerance specified in 3.6.2.

4.5.5 Backlash. With the mount positioned as specified in 4.6.2,. a torque of thirty pound inches shall be applied to the simulated gun trunnion shaft while viewing through collimator. The error if any when read on the collimator reticle shall be within the tolerance specified in 3.6.3. This. test shall be performed in both clockwise and counterclockwise direction.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with Specification MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P8237726. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The Mount, Periscope MI05 is primarily intended as a support and a drive for the direct fire periscope of the 106MM Multiple Rifle Self-Propelled Vehicle M60 (T165E2) (ONTOS Vehicle).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Selection of applicable levels of preservation, packaging and packing.

(c) Applicable stock number,

Custodian:
Army—MU

Preparing activity:
Army-MU
Project No. 1240-A374Y

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE 0-8279

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